Matthias Masiero

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PROFILE

Computer Science student at Santa Clara University with a strong foundation in coding and a passion for turning data into insights. Experienced in machine learning and artificial intelligence, predictive modeling, software development, AWS cloud architecture, and currently expanding into quantum computing through projects with IBM Quantum and Qiskit. Passionate about building scalable, intelligent systems that bridge theory with real-world impact.

EDUCATION

Bachelor of Science in Computer Science, Minor in Physics — 2027
Santa Clara University — Santa Clara, CA

SKILLS

Programming Languages: Python, Java, C++, Swift, HTML, CSS, JavaScript, SQL, TypeScript, OCaml, Qiskit, OpenQASM

Tools & Technologies: AWS (Lambda, EC2, S3, SageMaker, CloudWatch, API Gateway, Amplify), Docker, CI/CD Pipelines, TensorFlow, Bash, Git, GraphQL, IBM Quantum Platform, AI Agents (Claude Code, N8N, Amazon Q, LangChain, LlamaIndex)

PROJECTS

DEVANGEL – 2025 AWS HACKATHON WINNER (DEVOPS INCIDENT DASHBOARD)
- Built a real-time incident analysis platform using AWS CloudWatch Logs, Lambda, Step Functions, API Gateway, Bedrock, and EC2 to aggregate metrics and detect and respond to anomalies across production systems.
- Integrated an LLM-powered AI agent (Amazon Q for GitHub) that autonomously generates code-level fixes and opens pull requests when incident root causes are detected.
- Designed a scalable AWS architecture enabling autonomous debugging, intelligent alerting, and DevOps automation.

LOW-LATENCY ORDER MATCHING ENGINE
- Engineered high-performance order matching engine in C++ achieving 300K orders/sec with 3.3μs mean latency; profiled and optimized hot path for 30x speedup using cache-aligned structures, SPSC ring buffers, and zero-allocation design.
- Implemented price-time priority matching supporting limit, market, and stop orders with O(1) operations via intrusive linked lists and 1M-object pre-allocated pool.
- Built production-ready event-driven TCP server (kqueue/epoll) with non-blocking I/O, comprehensive input validation, and binary protocol; load tested 3M+ orders with zero data loss.

CARDTEMPO – CREDIT CARD OPTIMIZATION PLATFORM (2nd Place, SCU Winter Challenge)
- Built a profitable fintech platform improving credit scores by 15-160 points; full-stack Next.js/TypeScript (25K+ LOC) with Stripe subscriptions, Supabase PostgreSQL + RLS, Redis rate limiting, and defense-in-depth security.
- Engineered card recommendation engine using fuzzy string matching, multi-criteria optimization (6 weighted factors), and spending cap modeling across 11 categories with signup bonus attainability checks.
- Designed 4 payment allocation strategies with greedy threshold-crossing algorithm and multi-criteria scoring (utilization, APR, urgency, limit weights); built 6 what-if calculators with severity multiplier-based credit score modeling.

EXPERIENCE & LEADERSHIP

Sports Science Internship, *Santa Clara University Athletics Jan 2025 - Present | Santa Clara, CA*
- Processing 750K+ biometric data points per session across 22 athletes at 10Hz sampling frequency from Catapult GPS/accelerometer wearables to predict and prevent non-contact injuries.
- Engineering feature-extraction pipelines and applying regression and classification algorithms to optimize training loads.
- Collaborating with trainers and data scientists to refine predictive models for injury prevention and recovery, targeting a 70% reduction in non-contact injuries.

Machine Learning Researcher, *Santa Clara University Nov 2025 - Present | Santa Clara, CA*
- Benchmarked 6 transformer models (BERT, RoBERTa, MentalRoBERTa, ELECTRA, LLaMA-3.2) for suicidal ideation detection across 5 Reddit and Twitter datasets, achieving up to 98% accuracy on binary classification tasks.
- Designed a 40 sample adversarial test set spanning implicit ideation, sarcasm, and ambiguous distress to stress test model robustness, exposing failure modes where surface level lexical cues proved insufficient.

ACM Event Coordinator, *Santa Clara University Oct 2024 - Present | Santa Clara, CA*
- Board member of the largest computer science club on campus; lead AI and quantum computing workshops teaching machine learning fundamentals and Qiskit implementations.
- Plan and run two major hackathons each year, with over 350 participants and multiple guest speakers.